```
clc;
clear;
close all;

% Audio file ko load karna
[audio, Fs] = audioread('snighda.audio.wav.mp4');

% Stereo audio ko mono me convert karna
if size(audio,2) > 1
    audio = mean(audio,2);
end

% Audio ka time axis
t = (0:length(audio)-1)/Fs;

% Original audio ka graph
figure;
plot(t,audio);
xlabel('Time (seconds)');
ylabel('Amplitude');
title('Original Audio Signal');
grid on;

% Original audio play karna
sound(audio,Fs);
% T1 aur T2 ke time intervals
T1 = (t >= 1.2 & t <= 2.1);
T2 = (t >= 3.6 & t <= 4.4);

% T1 signal
figure;
plot(t(T1), audio(T1));
xlabel('Time (seconds)');
ylabel('Amplitude');
title('Signal T1');
grid on;

% T1 audio play
player1 = audioplayer(audio(T1), Fs);
playblocking(player1);

% T2 signal
figure;
plot(t(T2), audio(T2));
xlabel('Time (seconds)');
ylabel('Amplitude');
title('Signal T2');
grid on;

% T2 audio play
player2 = audioplayer(audio(T2), Fs);
playblocking(player2);
```

```
% 0.5 second advance ke liye samples calculate karna
shift = round(0.5 * Fs);

% Signal ko 0.5 second advance karna
y_advance = [audio(shift+1:end); zeros(shift,1)];

% Advance signal ka time axis
t_advance = (0:length(y_advance)-1)/Fs;

% Advance signal ka graph
figure;
plot(t_advance, y_advance);
xlabel('Time (seconds)');
ylabel('Amplitude');
title('Advance Signal y(t) = x(t + 0.5)');
grid on;

% Advance signal ko play karna
player3 = audioplayer(y_advance, Fs);
playblocking(player3);
% Time expansion by factor 10
t_scaled = t * 10;

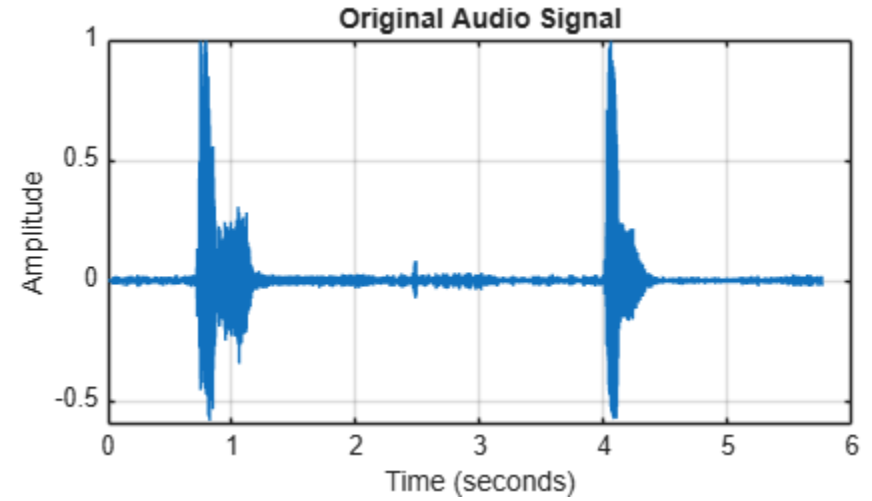
% Scaled signal ka graph
figure;
plot(t_scaled, audio);
xlabel('Time (seconds)');
ylabel('Amplitude');
title('Time Scaled Signal y(t) = x(t/10)');
grid on;

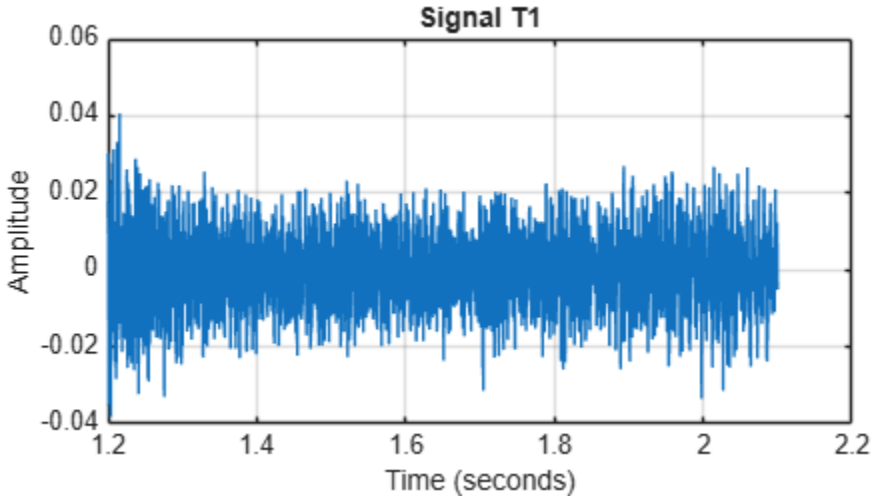
% Scaled signal ko slow speed par play karna
player4 = audioplayer(audio, Fs/10);
playblocking(player4);
% Time reversal / flip operation
y_flip = flip(audio);

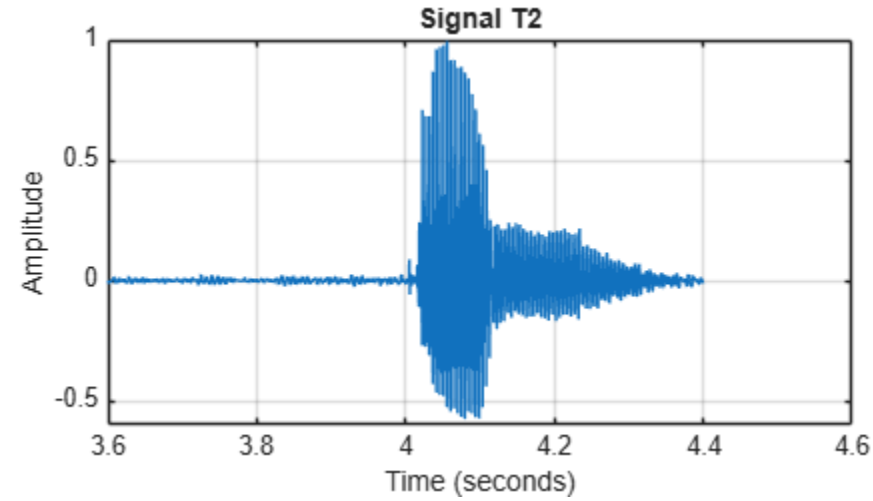
% Reversed time axis
t_flip = -fliplr(t);

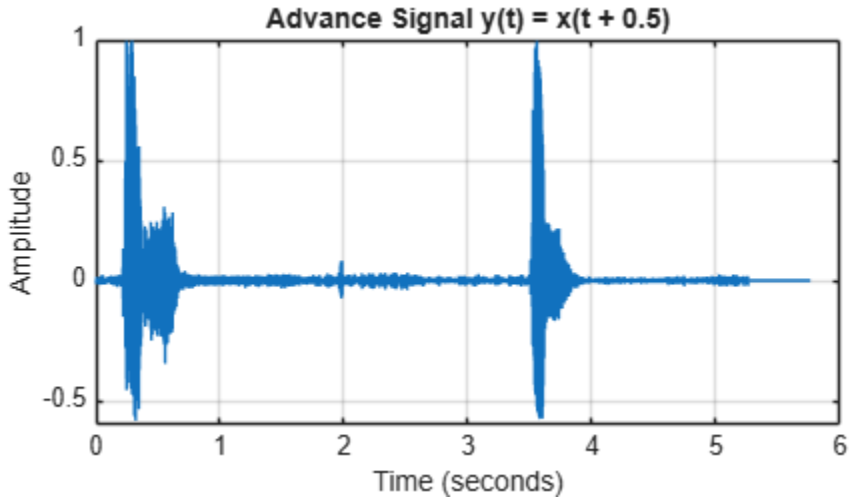
% Flipped signal ka graph
figure;
plot(t_flip, y_flip);
xlabel('Time (seconds)');
ylabel('Amplitude');
title('Flipped Signal y(t) = x(-t)');
grid on;

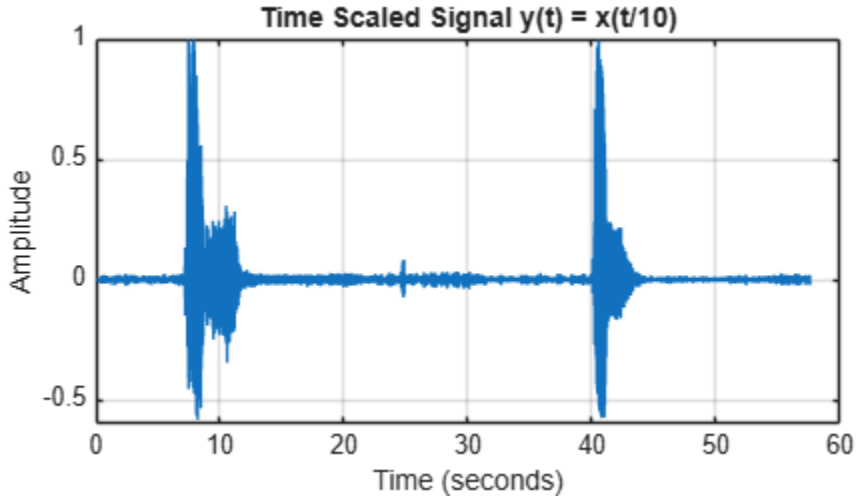
% Flipped audio ko play karna
player5 = audioplayer(y_flip, Fs);
playblocking(player5);
```

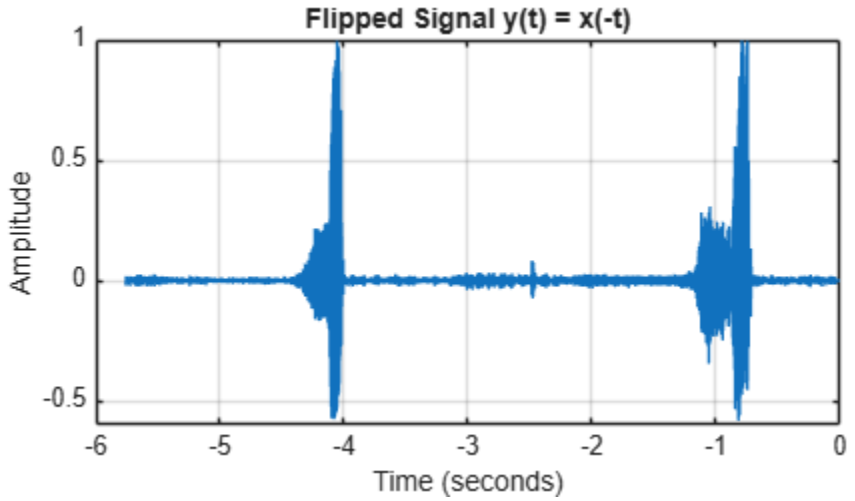

Original Audio Signal
Amplitude
Time (seconds)


Signal T1
Amplitude
Time (seconds)


Signal T2
Amplitude
Time (seconds)







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